

Ex. No.: 11b)  
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### LRU

#### Aim:

To write a c program to implement LRU page replacement algorithm.

#### Algorithm:

- 1: Start the process
- 2: Declare the size
- 3: Get the number of pages to be inserted
- 4: Get the value
- 5: Declare counter and stack
- 6: Select the least recently used page by counter value
- 7: Stack them according to the selection.
- 8: Display the values
- 9: Stop the process

#### Program Code:

```
#include <stdio.h>

int main() {
    int refstr[100], frames[20], recent[20];
    int refsize, framesize;
    int i, j, k, time = 0, pf = 0, isHd, index;
    printf("Enter the Number of pages: ");
    scanf("%d", &refsize);
    for (int i = 0; i < refsize; i++) {
        printf("[%d]: ", i + 1);
        scanf("%d", &refstr[i]);
    }
    printf("Enter page frame size: ");
    scanf("%d", &framesize);
    for (i = 0; i < framesize; i++) {
        frame[i] = -1;
        recent[i] = 69;
    }
```



```

printf("\n");
for(int i=0; i < refsize; i++){
    isHit = 0;
    for(int j=0; j < framesize; j++){
        if(frames[j] == refstr[i]){
            isHit = 1;
            recent[j] = time++;
            break;
        }
    }
}

```

```

if(isHit){
    printf("%d → No page fault\n", refstr[i]);
    continue;
}

```

```

int emptyind = -1;
for(j=0; j < framesize; j++){
    if(frames[j] == -1){
        emptyind = j;
        break;
    }
}

```

```

if(emptyind != -1){
    frames[emptyind] = refstr[i];
    recent[emptyind] = time++;
}

```

```

} else {
    int min = recent[0];
    int index = 0;
}

```



```
for (j=1; j < framesize; j++) {
```

```
    if (recent[j] < min) {
```

```
        min = recent[j];
```

```
        tindex = j;
```

```
frames[tindex] = refer[i];
```

```
recent[tindex] = time++;
```

```
}
```

```
Rf++;
```

```
printf ("%2d → ", refer[i]);
```

```
for (int k=0; k < framesize; k++) {
```

```
    if (frames[k] != -1)
```

```
        printf ("%d", frames[k]);
```

```
    }
```

```
printf (" ⇒ page fault\n");
```

```
}
```

```
printf ("In Total page faults : %d\n", Rf);
```

```
}
```



OUTPUT:

Enter number of pages : 14

Enter [1] = 7

Enter [2] = 0

Enter [3] = 1

Enter [4] = 2

Enter [5] = 0

Enter [6] = 3

Enter [7] = 0

Enter [8] = 4

Enter [9] = 2

Enter [10] = 3

Enter [11] = 0

Enter [12] = 3

Enter [13] = 2

Enter [14] = 3

Enter page frame : 4

7  $\rightarrow$  7  $\Rightarrow$  page fault

0  $\rightarrow$  7 0  $\Rightarrow$  page fault

1  $\rightarrow$  7 0 1  $\Rightarrow$  page fault

2  $\rightarrow$  7 0 1 2  $\Rightarrow$  page fault

0  $\rightarrow$  NO page fault

4  $\rightarrow$  3 0 4 2  $\Rightarrow$  page fault

2  $\rightarrow$  NO page fault

3  $\rightarrow$  NO page fault

0  $\rightarrow$  NO page fault

3  $\rightarrow$  NO page fault

2  $\rightarrow$  NO page fault

3  $\rightarrow$  NO page fault

Total page faults : 6.

**Sample Output :**

Enter number of frames: 3

Enter number of pages: 6

Enter reference string: 5 7 5 6 7 3

5 -1 -1

5 7 -1

5 7 -1

5 7 6

5 7 6

3 7 6

Total Page Faults = 4

✓ Q. 11

**Result:**

A C program for finding the page fault using LRU page replacement technique is implemented successfully.